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Swing-Up and Stabilization of a Cart-Pole with a Four-Neuron Analog Spiking Network

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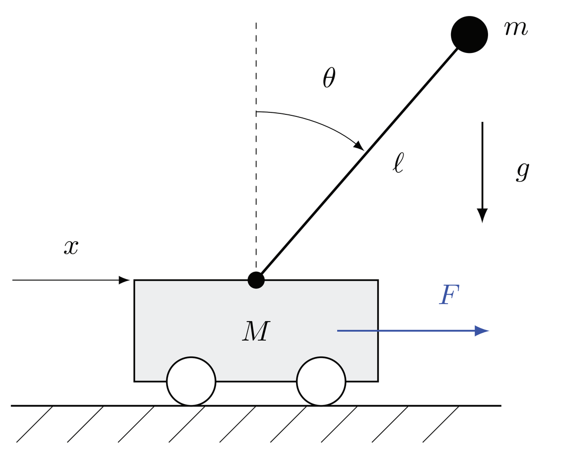
MOTIVATION / RESEARCH QUESTION

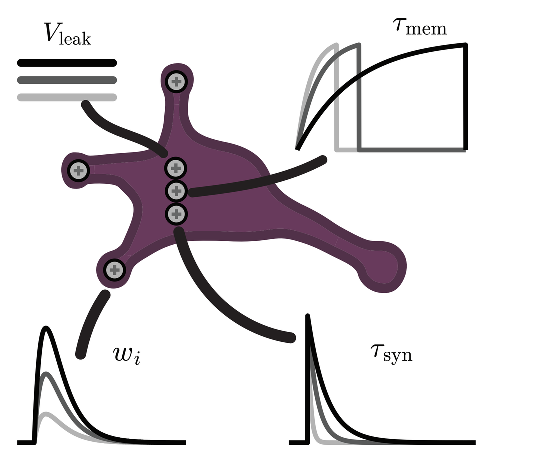
Leaky integrate-and-fire (LIF) neurons offer an energy-efficient route to neuromorphic control [1, 2]. This work tests Lu.i, a palm-sized analog LIF circuit [3], on the nonlinear cart-pole benchmark. Unlike ideal simulations of these neurons, physical neurons introduce latency, noise, mismatch, and limited weight precision. Rather than using large populations for accurate rate coding, as in near-linear LIF control [4], we use thresholded latency responses as sign activations in a minimal controller.

Research Question

- Can four analog Lu.i neurons implement a sign-activation controller for cart-pole swing-up and stabilization?
- How reliable is the transferred physical controller across repeated trials and initial offsets?
- What calibration information is needed to reproduce transfer from simulation to noisy analog hardware?

SYSTEM OVERVIEW / THEORY





Cart-Pole

The plant is the standard simulated cart-pole [5, 6]: a pole of mass m and length ℓ hinged to a cart of mass M. The upright state is $\theta = 0$; the hanging rest state is $\theta = \pm\pi$. The controller reads only $(\theta, \dot\theta, \dot x)$, not cart position, and applies bang-bang actuation $F = F_{\max} z_{\text{out}}$, where $z_{\text{out}} \in \{-1, 1\}$ is the controllers output.

Lu.i Neuron

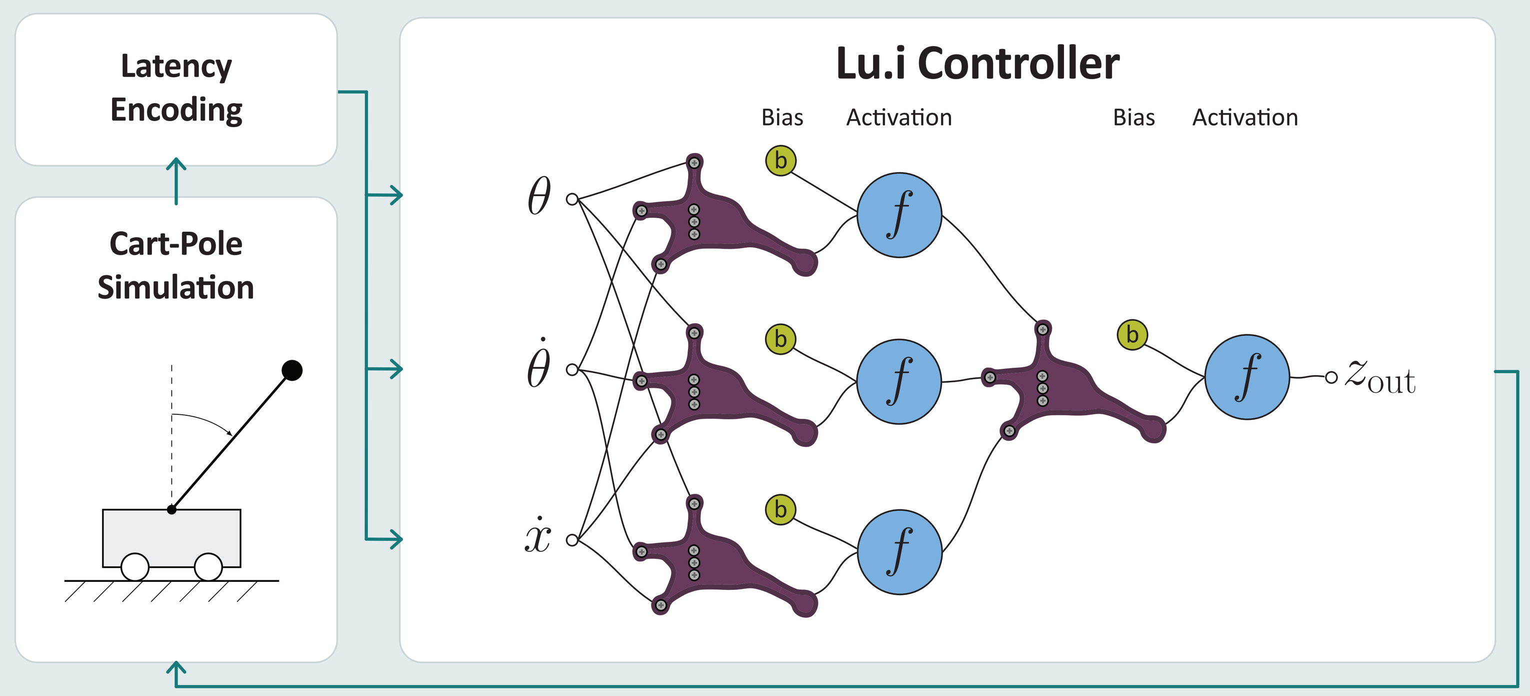
Each neuron is a palm-sized analog leaky integrate-and-fire (LIF) cell [3]: current-based synaptic inputs are integrated on a leaky membrane, and when the membrane crosses its threshold the neuron fires a spike and resets. Its three synapses have potentiometer-set weights with selectable sign (excitatory/inhibitory).

METHODS

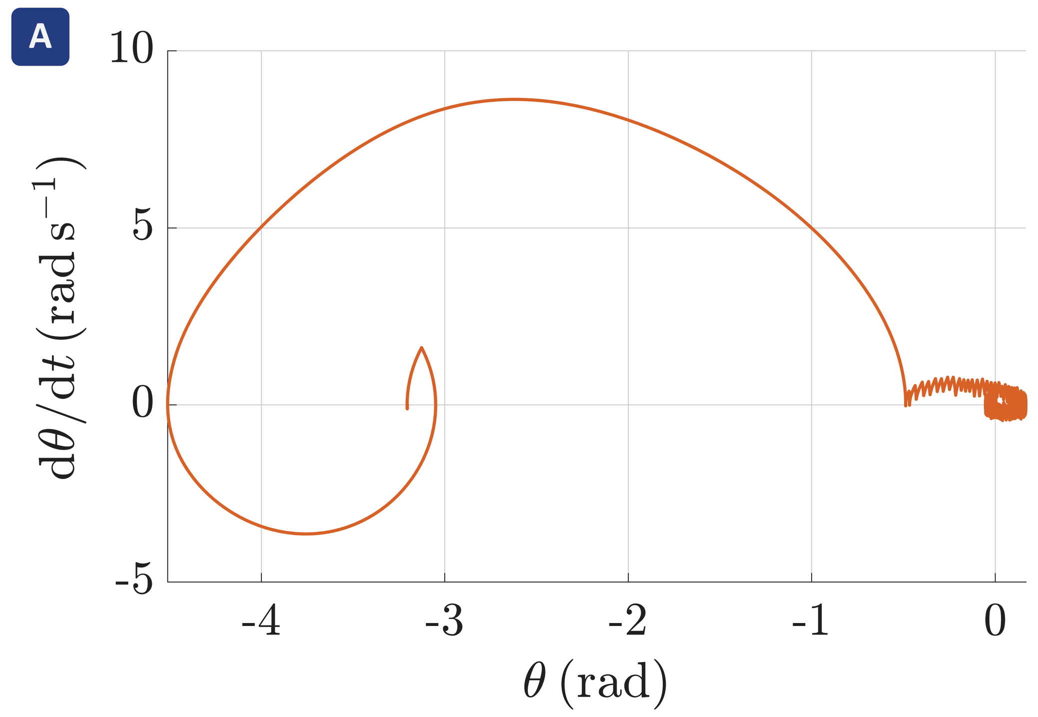
- **Cart-pole Simulator**
  The cart-pole equations were integrated with a custom RK4 simulator using $\Delta t = 1/60\,\text{s}$, $t_f = 60\,\text{s}$, and $F_{\max} = 8\,\text{N}$.
- **Controller.** The controller is a compact 3-3-1 fully connected feed-forward sign activated network [7]: three latency-encoded inputs, three hidden Lu.i neurons, and one output Lu.i neuron.
- **Core contribution: sim-to-analog transfer.**
  1. **Train a neural network in simulation.** An identical ANN was optimized offline with a gradient-free evolutionary strategy [8] and an energy-based swing-up fitness [9].
  2. **Encode state variables as spike rates.** The simulated state variables $z_{\text{in},i} \in \{\theta, \dot\theta, \dot x\}$ were mapped to input frequencies:
     $$f_{\text{in},i} = f_{\text{offset},i} + f_{\text{scale},i} z_{\text{in},i}$$
  3. **Approximate local analog spike-rate transfer.** Around a measured baseline, the Lu.i response was approximated by a first-order spike-rate model:
     $$f_{\text{out}} \approx f_{\text{baseline}} + \sum_i w_i (f_{\text{in},i} - f_{\text{offset},i}).$$
     Here, w_i denotes an effective rate-transfer coefficient, not the raw potentiometer setting.
  4. **Transfer trained weights to hardware.** The trained weights were implemented by tuning Lu.i potentiometers. Baseline firing rates were measured at start-up, and biases were applied digitally through an Arduino.
  5. **Threshold the physical output rate into a force command.** The output firing rate was estimated from the inter-spike interval Δt_{spike}, $f_{\text{out}} = \frac{1}{\Delta t_{\text{spike}}}$, and thresholded to produce the bang-bang control signal:
     $$z_{\text{out}} = \operatorname{sign}(f_{\text{out}} - f_{\text{baseline}} + f_{\text{bias}})$$
     Only the 3 hidden neurons and 1 output neuron are physical Lu.i neurons; the 3 inputs are externally encoded by spike-rate.
- **Evaluation metrics.** A trial was counted as successful if the pole reached the upright neighborhood $|\theta(t)| < \epsilon$ for at least $12\,\text{s}$. The settling time t_{stab} was defined as the first time after which the trajectory remained inside this region for the full hold interval.

CONTROLLER ARCHITECTURE

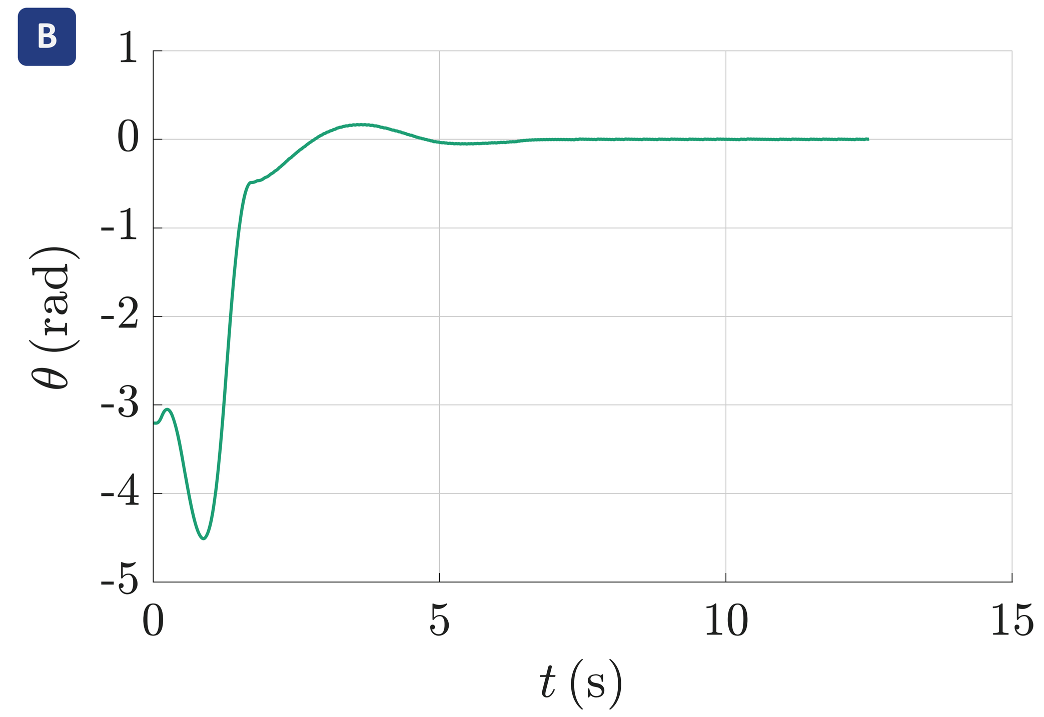
Pipeline from sensing to actuation



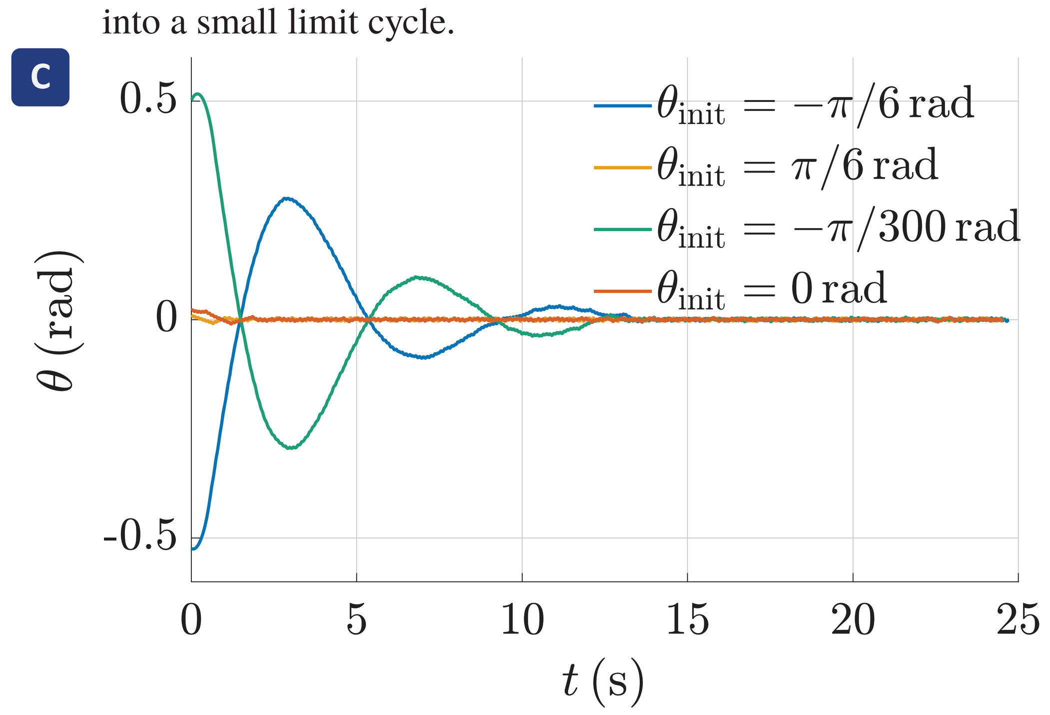
RESULTS



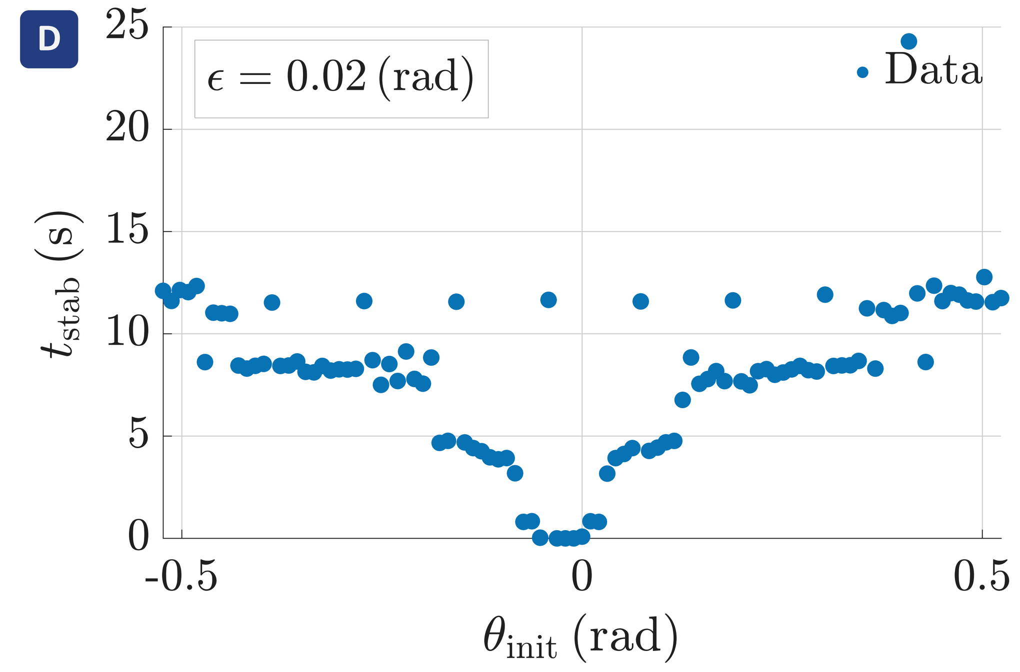
(a) Phase portrait for swing-up from the hanging state. The trajectory approaches the upright equilibrium and settles into a small limit cycle.



(b) Pole angle during swing-up. The pole reaches the upright neighborhood after approximately $5\,\text{s}$.



(c) Stabilization from selected initial offsets near the upright equilibrium.



(d) Settling time as a function of initial pole angle, using the threshold $\epsilon = 0.02\,\text{rad}$.

Closed-loop behavior of the physical Lu.i controller in hardware-in-the-loop experiments: (a)–(b) swing-up from the hanging state and (c)–(d) stabilization from small initial offsets.

In hardware-in-the-loop experiments, the physical Lu.i controller swung the pole from hanging to upright in approximately $5\,\text{s}$. After capture, the system remained near upright in a small bang-bang limit cycle of about $\pm 0.01\,\text{rad}$. From local initial offsets, the controller stabilized the pole for $|\theta_{\text{init}}| \lesssim 0.524\,\text{rad}$, beyond that, the pendulum falls over. Settling time increases from approximately under 1 to $11\,\text{s}$ with increasing $|\theta_{\text{init}}|$. Across 101 repeated hardware trials, 98 were successful, giving 97.03% empirical success.

DISCUSSION / LIMITATIONS

The results show that a four-neuron physical analog controller can solve both swing-up and local stabilization in a simulated cart-pole loop. The observed limit cycle is expected: sign activations and fixed-magnitude force cannot produce arbitrarily small corrective inputs. Finite spike-counting windows, analog latency, and device noise likely contribute to residual jitter and settling-time scatter.

Limitations & Outlook

- The plant was simulated, so the experiment demonstrates hardware-in-the-loop control rather than control of a mechanical cart-pole.
- Future work should quantify ideal simulation, simulation with measured calibration error, and physical Lu.i performance.
- Because cart position x is not sensed, future evaluations should report cart drift and include rail constraints.
- Biasing and parts of thresholding remain digital; a fuller analog implementation should quantify or reduce these computations.

KEY RESULT

The four-neuron Lu.i controller swung up and stabilized the cart-pole in 98/101 hardware-in-the-loop trials, achieving a 97% success rate.

CONCLUSION / OUTLOOK

A compact 3-3-1 neural controller implemented with four Lu.i neurons can swing up and stabilize a simulated cart-pole using only $(\theta, \dot\theta, \dot x)$. The $98/101$ success rate supports the feasibility of low-cost analog LIF hardware for nonlinear feedback control, while motivating more detailed calibration and simulation-to-hardware transfer analysis.

REPRODUCIBILITY

Code, full article, trained weights, biases and figures are all available on GitHub.
https://github.com/DrErty/ERP-Neural-Network

SCAN ME

ERP Poster
UTRECHT UNIVERSITY / FACULTY OF SCIENCE / SCM&B RESEARCH GROUP

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